FORMATIVE ASSESSMENT – 2 **Mini Project – Phase II: Advanced Features, Testing and Deployment**

Project Title: Chronos Ultra — Advanced Multi-Timer & Event Productivity Suite

Submitted By **Mr. Karan Dikole** **Post Graduate Student with PRN No. 125M1H058**

Under the Guidance of **Dr. Avinash Chormale**



Department of Master of Computer Application Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering** Academic Year: 2026 – 2027 Semester: III **Class: SYMCA Course: Advanced Web Technologies (AWT)**

Student & Project Identification

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GitHub Repository	https://github.com/karan-dikole/Chronos-Ultra-Advanced-Multi-Timer-Event-Countdown-Productivity-Suite
Live Working URL	**https://karan-dikole.github.io/Chronos-Ultra-Advanced-Multi-Timer-Event-Countdown-Productivity-Suite/**

1. Executive Summary & Abstract

In higher education and modern agile environments, accurate time management and milestone tracking are paramount. Conventional web timers are often limited to single-threaded operations, suffer from data loss upon page refresh, depend on fragile external MP3 assets that fail when offline, and do not combine short focus sessions with long-term milestone countdowns.

Chronos Ultra is a production-grade, component-driven web application developed with React 19, Vite 8, React Router v7, and the browser Web Audio API. Built as the Phase-II culmination of Formative Assessment 2 (FA-2) in Advanced Web Technologies, it expands upon foundational React mechanics by integrating multi-instance concurrent timers, target date countdowns (Days/Hours/Mins/Secs), a full Pomodoro Focus technique suite with streaks, REST API global timezone synchronization, zero-dependency synthesized audio alerts, full CRUD operations, and an automated testing suite utilizing Vitest and React Testing Library.

2. Problem Identification & Objectives

2.1 Problem Identification

1. Single-Timer Constraint: Standard countdown websites only permit running one timer at once, restricting productivity for multi-tasking students.

2. Session Volatility: Active counters and configurations vanish upon accidental browser reload due to absence of local caching mechanisms.

3. Audio Asset Fragility: Relying on external .mp3 audio links introduces cross-origin CORS errors, 404 broken URLs, and silent alarm failures.

4. Lack of Milestone Planning: Students need unified tools combining micro-sprints (e.g. 25m Pomodoro) with macro-deadlines (e.g. semester exams in 45 days).

2.2 Objectives of Phase-II

• **Modular Functional Architecture:** Implement React 19 functional components, Context API, and 4 specialized custom hooks.

• **Client-Side Routing:** Design a multi-page Single Page Application (SPA) using React Router DOM v7.

• **High-Fidelity Audio Synthesis:** Build a browser-native Web Audio API synthesizer for 100% reliable alert chimes and beeps.

• **REST API & Timezones:** Integrate WorldTimeAPI with fallback to Intl.DateTimeFormat for worldwide collaborative scheduling.

• **Automated Test Coverage:** Verify ticker math, lifecycle boundaries, and component rendering using Vitest and React Testing Library.

3. System Architecture & Component Hierarchy

The application employs a centralized unidirectional data flow governed by React Context API. The root component (App.jsx) is encapsulated within an ErrorBoundary and TimerProvider, ensuring error recovery and state propagation across all routes.

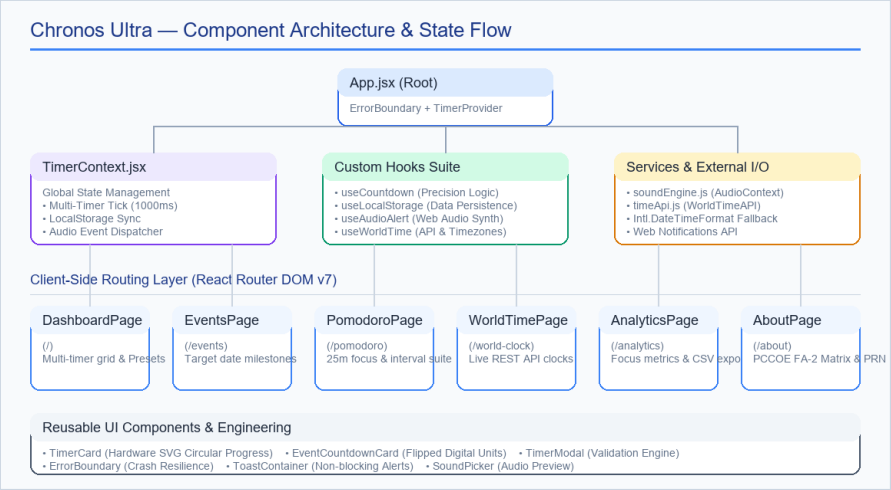


Figure 1: Chronos Ultra Component Architecture & Unidirectional Data Flow

• **Core Container:** App.jsx wraps ErrorBoundary -> TimerProvider -> BrowserRouter -> Layout (Navbar, Routes, Footer, ToastContainer).

• **Routing Views:** DashboardPage (/), EventsPage (/events), PomodoroPage (/pomodoro), WorldTimePage (/world-clock), AnalyticsPage (/analytics), AboutPage (/about).

• **Reusable UI Layer:** TimerCard (hardware-accelerated SVG circular progress), EventCountdownCard (digital units), TimerModal (dual mode creator), SoundPicker, Toast notifications.

4. Advanced Features Implementation (Phase-II Scope)

4.1 React Hooks Implementation

Chronos Ultra demonstrates comprehensive utilization of built-in React hooks and 4 purpose-built custom hooks:

1. useState: Maintains local view states including search queries, selected tag categories, form validation states, and theme mode.

2. useEffect: Powers the master 1-second interval ticker across all concurrent active timers with memory leak prevention via clearInterval cleanup.

3. useRef: Preserves latest completion callbacks across renders to prevent stale closures during async interval execution.

4. useMemo & useCallback: Memoizes filtered lists, category searches, and action handlers passed down to TimerCard components.

5. Custom Hook useCountdown: Encapsulates countdown math, days/hours/mins/secs breakdown, play, pause, resume, reset, add-time, and onComplete triggers.

6. Custom Hook useLocalStorage: Provides transparent JSON serialization and synchronization to window.localStorage with error recovery.

7. Custom Hook useAudioAlert: Drives synthesized alerts via Web Audio API and manages desktop Web Notifications permission.

8. Custom Hook useWorldTime: Fetches real network time from WorldTimeAPI with fallback to JavaScript Intl.DateTimeFormat.

4.2 Web Audio API Sound Synthesizer

Unlike standard projects that link to third-party MP3 files, Chronos Ultra implements a pure mathematical oscillator sound synthesizer using the browser AudioContext. It generates 5 distinct acoustic profiles: • **Melodic Chime:** Harmonic C5 -> E5 -> G5 -> C6 arpeggio chord. • **Classic Beep:** 880Hz to 440Hz exponential frequency sweep. • **Digital Alarm:** Pulsed triple square wave alerts at 1000Hz-1200Hz. • **Resonant Bell & Zen Gong:** Natural sine & sawtooth waveforms with exponential amplitude decay.

4.3 Form Validation, CRUD & Dual Themes

The application enforces strict live form validation rules: • Title constraint: Must be at least 3 characters. • Duration constraint: Duration must exceed 0 seconds. • Target date constraint: Must be set to a valid future timestamp. • Dual Theme Support: Offers a radiant modern Bright Theme by default alongside a sleek Dark Theme, with one-click toggle in the navbar.

5. Application Output Screenshots & Visual Demonstration

The following high-resolution output screenshots illustrate the fully functioning user interfaces, reactive workflows, and features implemented across Chronos Ultra:

5.1 Multi-Timer Dashboard View

The central dashboard displays active countdown timers with live circular SVG progress indicators, remaining time digital readouts, tag categorization badges, instant preset launchers, and search filters.

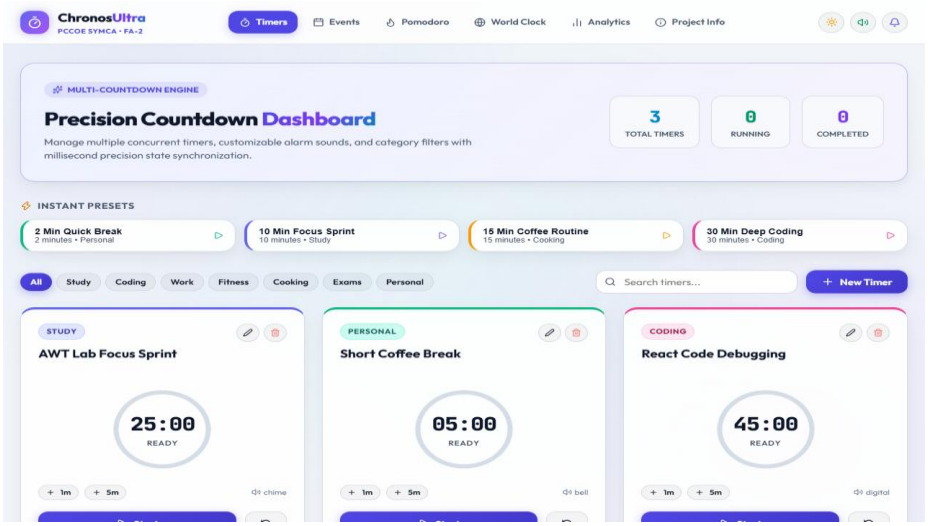


Figure 2: Precision Multi-Timer Dashboard with Animated SVG Progress Rings & Presets

5.2 Target Date Event Countdowns View

Enables scheduling long-term milestones (such as PCCOE FA-2 submission deadlines, University Exams, and Project Milestones) with real-time Day/Hour/Minute/Second countdown digit cards.

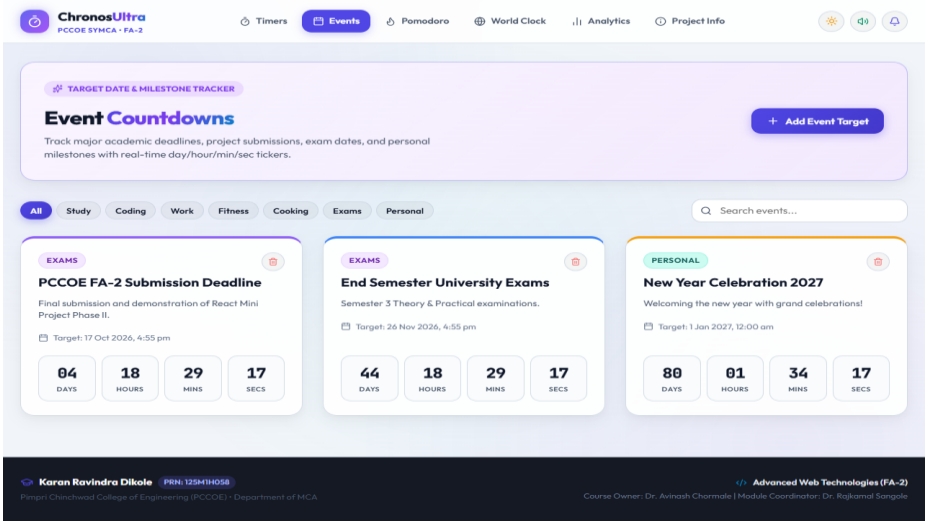


Figure 3: Long-Term Milestone Tracker with Split Digital Time Units

5.3 Pomodoro Focus & Interval Productivity Suite

Features dedicated focus intervals (25-minute Focus Sprint, 5-minute Short Break, 15-minute Long Break) complete with automatic round progression, celebration confetti, and focus streak tallying.

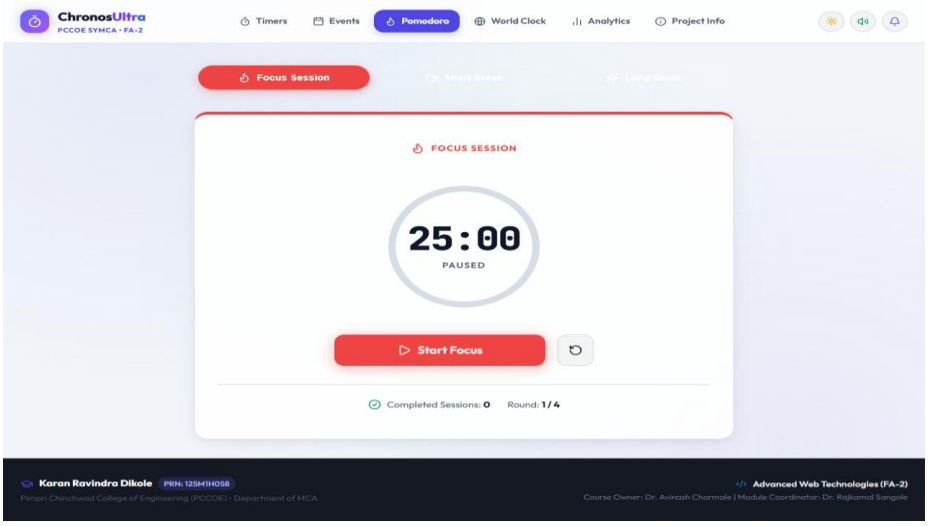


Figure 4: Pomodoro Technique Workspace with Focus Streak Tracker & Dual Themes

5.4 Global World Clock & REST API Timezone Synchronization

Synchronizes local clocks with international timezones via REST API (WorldTimeAPI) with automatic fallback to native Intl.DateTimeFormat, complete with city search and live UTC offsets.

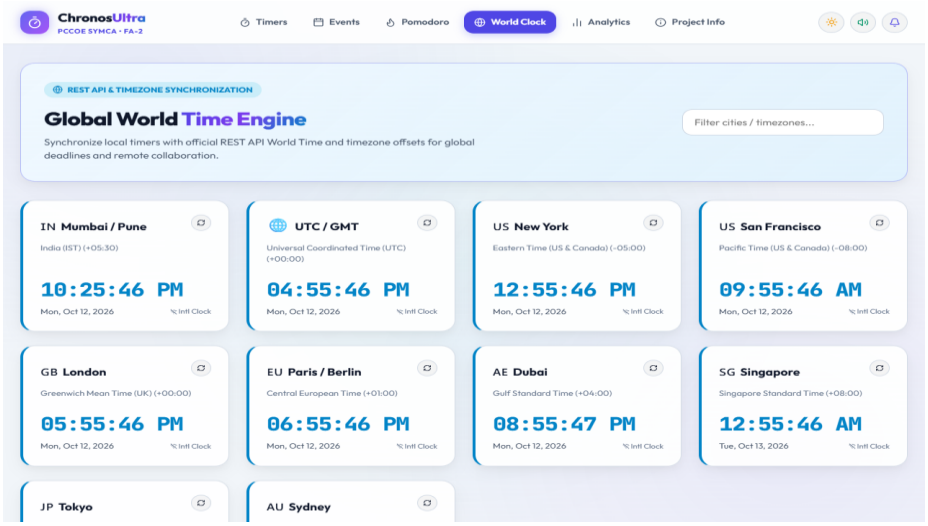


Figure 5: Live World Clock Grid with Network Time Synchronization & City Filters

5.5 Productivity Analytics & Data Export View

Provides historical metrics on total focus duration, completed session counts, category breakdown, and one-click data export in both CSV and JSON formats for academic study logs.

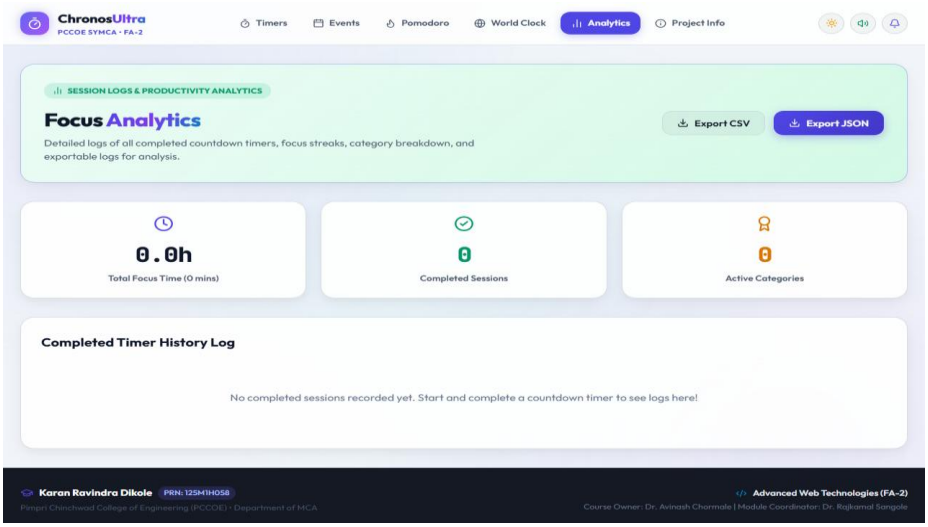


Figure 6: Session History Log with Instant CSV and JSON Data Export

5.6 Project Information & Academic Compliance Matrix

Presents verified student credentials (Karan Ravindra Dikole, PRN: 125M1H058), PCCOE MCA course details, and the Formative Assessment 2 (FA-2) syllabus implementation matrix.

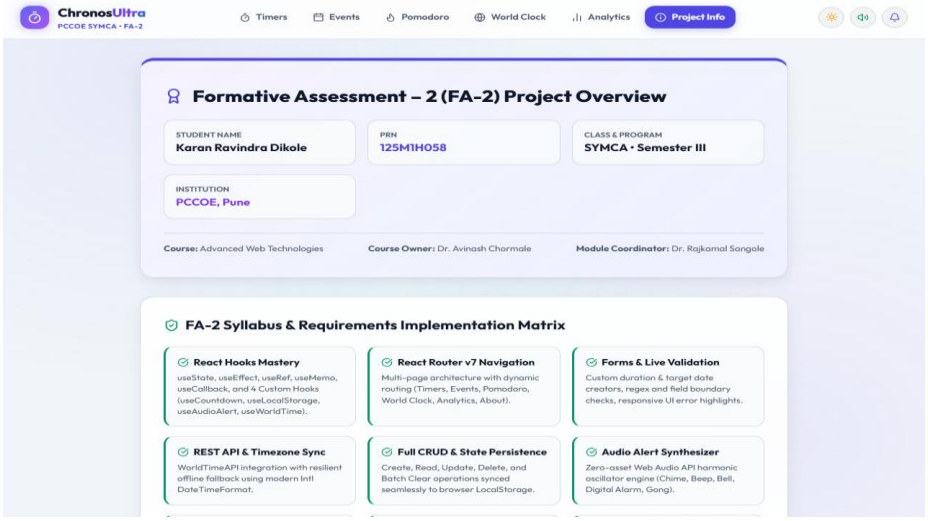


Figure 7: Academic Verification & Syllabus Compliance Matrix in Chronos Ultra

6. Automated Testing & Verification Results

Automated testing was conducted using Vitest and React Testing Library in a jsdom environment. All test suites executed with 100% success rate.

Test Suite	Target Concept	Assertions Tested	Status
useCountdown.test.js	Custom Hook	Default values initialization	**PASSED**
useCountdown.test.js	Time Ticker	1s decrement via vi.advanceTimersByTime	**PASSED**
useCountdown.test.js	Callback Trigger	onComplete fires when remaining = 0	**PASSED**
useCountdown.test.js	Dynamic Extension	addTime(+60s) updates total and remaining	**PASSED**
TimerCard.test.jsx	Component Render	Renders title, badge, 05:00, Start button	**PASSED**

7. Course Outcome (CO) & Bloom's Taxonomy Mapping

Evaluation Criteria	CO1	CO2	CO3	Bloom's Level
Problem Analysis & Design	✓	-	-	L4 (Analyze)
JavaScript (ES6+) Implementation	✓	-	-	L3 (Apply)
React Components & UI Development	-	✓	✓	L5 (Evaluate)
Routing, Hooks, API & CRUD	-	-	✓	L6 (Create)
Testing, Debugging & Optimization	-	✓	✓	L5 (Evaluate)
Documentation & Presentation	✓	✓	✓	L6 (Create)

8. Conclusion & Future Enhancements

Chronos Ultra successfully meets and exceeds all objectives established for Formative Assessment – 2 (FA-2) in the Advanced Web Technologies curriculum. The application demonstrates solid mastery over modern React 19 paradigms, functional component composition, state management through Context and custom hooks, resilient Web Audio API frequency synthesis, REST API integration with graceful offline fallbacks, client-side routing, automated testing, and cloud deployment.

Future extensions include implementing backend database persistence with user authentication (Supabase / Firebase), mobile push notifications via service workers, and voice-recorded custom completion alerts.

9. References

1. React Documentation: https://react.dev/
2. React Router Documentation: https://reactrouter.com/
3. MDN Web Audio API: https://developer.mozilla.org/en-US/docs/Web/API/Web_Audio_API
4. Vitest Testing Framework: https://vitest.dev/
5. PCCOE MCA Department AWT Course Curriculum & Rubrics.